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Synthesis of Chiral Tertiary Alcohols by Cu^{I}-Catalyzed Enantioselective Addition of Organomagnesium Reagents to Ketones

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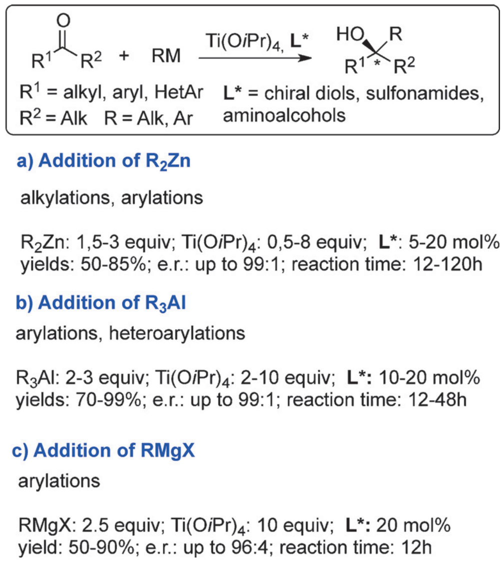
Abstract: Catalytic enantioselective addition of organometallic nucleophiles to ketones is among the most straightforward approaches to the synthesis of chiral tertiary alcohols. The first such catalytic methodologies using the highly reactive organomagnesium reagents, which are the preferred organometallic reagents in terms of cost, availability, atom efficiency, and structural diversity, were developed only during the last five years. This Concept article highlights the fundamental breakthrough that made the development of methodologies for highly enantioselective Cu^{I}-catalyzed alkylation of ketones using organomagnesium reagents possible.

Introduction

The central role of molecular chirality in nature places the availability of chiral products with high enantiomeric purity as a core objective of modern synthetic organic chemistry.[1] Chiral alcohols constitute a key class of organic molecules, due to their widespread occurrence in natural products and pharmaceuticals, as well as their application as chiral building blocks in synthetic chemistry.[2] Catalytic asymmetric carbon–carbon bond formation, which is at the heart of chemical synthesis, represents the most straightforward approach for the preparation of single enantiomers of chiral secondary and tertiary alcohols.[1] Addition of organometallics to carbonyl compounds is one of the most direct methods to achieve this.[1,3]

Research over the last century has established a range of organometallic reagents available to organic chemists for this purpose.[3] The most commonly used organometallics are organozinc (R_2Zn) and organoaluminum (R_3Al) reagents (Scheme 1 a, b). Despite their cost, structural limitations, long reaction times, and the fact that they transfer only one of the alkyl groups, organozinc reagents were considered as the reagents of choice due to the ability to tune their reactivity by using additives. In particular, the catalytic asymmetric addition of organozinc reagents to aldehydes has been well developed over the past fifty years. In contrast, the catalytic asymmetric addition of organometallic reagents to ketones is more difficult, primarily due to the decreased reactivity of ketones compared to aldehydes and decreased enantiodiscrimination due to the smaller steric and electronic differences between the two substituents at the carbonyl group.[3] Recent advances in this field have enabled the catalytic asymmetric addition of organozinc reagents to ketones, accessing enantioenriched tertiary alcohols in moderate to high yields with excellent enantioselectivities (Scheme 1 a).[3] Dosa and Fu reported the first catalytic asymmetric example of the addition of diphenylzinc to ketones.[4] Shortly thereafter, Yus and co-workers reported the first catalytic asymmetric addition of dialkylzinc reagents.[5] These initial results inspired the design of better ligands for the asymmetric addition of dialkylzinc reagents to ketones and led to an increase in substrate scope.[6,7] In addition to organozinc reagents, several methodologies that use organoaluminum reagents have also been developed (Scheme 1 b).[8]



Scheme 1. Titanium-promoted catalytic asymmetric alkylations/arylations of ketones.[3g]

Importantly, the most desirable organometallics in terms of cost, availability, atom efficiency, and structural diversity are the highly reactive organomagnesium reagents (Grignard reagents). However, direct catalytic asymmetric additions to both aldehydes and ketones with these organometallics were not realized until recently, mainly due to the challenge of controlling the following characteristics: i) Their extreme reactivity profile, which results in uncatalyzed reactions; ii) their basicity, leading to enolization; and iii) competitive reduction through β-H transfer. A first step towards this was taken in 2008 when the group of Harada reported that highly enantioselective catalytic addition of Grignard reagents to aldehydes is possible, albeit requiring the use of excess titanium tetraisopropoxide.[9] Subsequently the group of Yus and Maciá reported in 2014 that addition of aryl Grignard reagents to ketones is also feasible (Scheme 1 c).[10]

A common thread through all of these reactions based on using Zn, Al, and Mg reagents is the need to employ superstoichiometric amounts of organometallic reagents (up to 3 equiv) as well as titanium tetraisopropoxide (up to 10 equiv) to obtain good yields and enantiomeric ratios.[3g,11] The accepted view is that in these reactions organotitanium species or titanates are formed, which act as the actual nucleophile. Although organotitanium species are less reactive and less basic than organomagnesium reagents, they are nevertheless more nucleophilic than dialkylzinc reagents.[3g]

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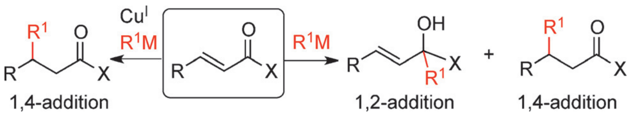
Separately from the developments of titanium-promoted addition of Zn, Al, and Mg reagents to carbonyl compounds, the development of methodologies for the addition of the same reagents to the α,β-unsaturated subclass of carbonyl compounds has centered on promotion by Cu^{I} salts.[12] The crucial distinction between these systems lies in the selectivity. A key challenge to the use of Zn-, Al-, and Mg-based reagents in addition reactions to α,β-unsaturated carbonyl compounds is the regioselectivity, that is, 1,4-addition (conjugate addition) versus 1,2-addition (direct addition). In the absence of any catalyst, the tendency of hard organometallics is to give a racemic mixture of 1,2- and 1,4-addition products with a preference for the former (Scheme 2).



Scheme 2. 1,4- versus 1,2-addition of hard organometallic reagents to α,β-unsaturated carbonyl compounds.

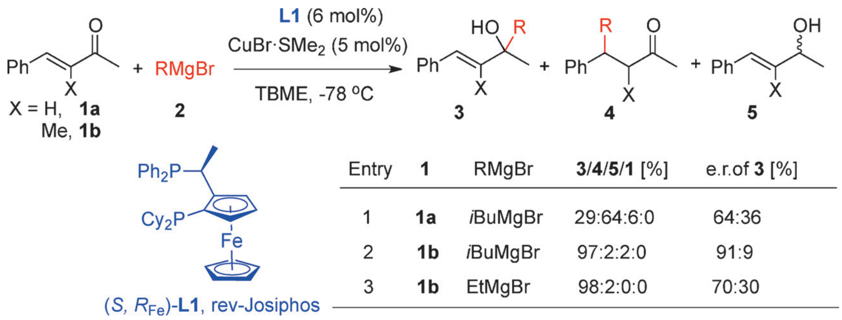
Whereas titanium-promoted reactions follow preferentially the 1,2-addition pathway, in the presence of by Cu^{I}-based catalysts the 1,2-addition is outcompeted by steering the selectivity towards 1,4-addition.[12] This has been established since 1941 when Kharash and Tawney[13a] discovered that it was possible to shift the regioselectivity of the addition of Grignard reagents to α,β-unsaturated ketones through the presence of a Cu^{I} salt. Since then, Cu^{I}-based catalysts have become a mainstay in direct 1,4-additions, which led, over the following seventy years, to the development of many catalytic asymmetric 1,4-addition methodologies using organozinc, organoaluminum, and organomagnesium reagents.[12,14] As a consequence, Cu^{I} complexes have been overlooked as catalysts in the alkylation of ketones and aldehydes with hard organometallic reagents.[15]

A major step forward was made in recent years by the groups of Harutyunyan and Minnaard, when Cu^{I}-based chiral catalysts proved useful for the synthesis of enantiopure tertiary alcohols.[16] This discovery allowed, for the first time, the direct use of highly reactive and cost-efficient Grignard reagents in catalytic asymmetric alkylations of enolizable ketones, a rare feat and therefore an important discovery. The corresponding chiral tertiary alcohols were obtained with excellent yields in short reaction times and enantiomeric ratios of up to 99:1. Moreover, no additives to transform Grignard reagents into organotitanium or excess reagents were required. Furthermore, it was shown that the overwhelming 1,4-selectivity of Cu^{I}-based chiral catalysts in the addition of Grignard reagents to α,β-unsaturated ketones can be shifted towards the 1,2-addition product.[16a,b]

In this Concept article, the journey that led to the discovery of Cu^{I}-catalyzed asymmetric synthesis of chiral tertiary alcohols with Grignard reagents is reviewed, and the scope of this methodology, the possible mechanism, and its implications for further research are discussed.

Discovery: 1,4- versus 1,2-Selectivity

In the absence of any catalyst, the addition of Grignard reagents to α,β-unsaturated ketones typically can proceed through four different reaction pathways, namely: i) 1,2-addition to form product **3**; ii) 1,4-addition to form product **4**; iii) β-hydride transfer from Grignard reagents bearing β-hydride to form product **5**; iv) enolization of the initial ketone (Scheme 3). In the presence of Cu^{I} salt, however, 1,4-addition is strongly favored over the competing pathways. This has led to the development of catalytic enantioselective methodologies using Cu^{I} complexes containing chiral ligands.[12,14] Specifically, in an extensive body of work from Feringa and co-workers, Cu^{I} catalysts were utilized to exploit the 1,4-addition of Grignard reagents to a number of conjugated carbonyl compounds, including cyclic and acyclic ketones.[14d–g] Although in all documented reactions a certain fraction of 1,2-addition product is also formed, typically in 5–20% yield, this product was found to be racemic, and the general view was that this was the result of non-catalytic direct addition of Grignard reagents.[14d,g]



Entry	1	RMgBr	3/4/5/1 [%]	e.r.of 3 [%]
1	**1a**	*i*BuMgBr	29:64:6:0	64:36
2	**1b**	*i*BuMgBr	97:2:2:0	91:9
3	**1b**	EtMgBr	98:2:0:0	70:30

Scheme 3. Optimization data for the addition of Grignard reagents to α,β-unsaturated ketones.[16a, 18]

The origin of the 1,2-addition product was revisited, inspired by intriguing results on a different type of reaction, namely copper catalyzed C–H bond formation. A report by Lipshutz et al[17] demonstrated that in Cu^{I}-catalyzed asymmetric reduction of α,β-unsaturated ketones the traditional preference for 1,4-selectivity could be shifted to 1,2-addition instead.

The groups of Harutyunyan and Minnaard started by carrying out addition of Grignard reagents to α-H-substituted α,β-unsaturated ketone (**1a**) in the same conditions (−78 °C, in TBME) as previously reported by the Feringa group.[14d,g] Two reactions were carried out, one without any catalyst (no Cu^{I} salt or chiral ligand) and one in the presence of 5 mol% of Cu^{I} salt, but no chiral ligand.[16a] Comparing these reactions, it was found that in the presence of Cu^{I} salt the amounts of both the

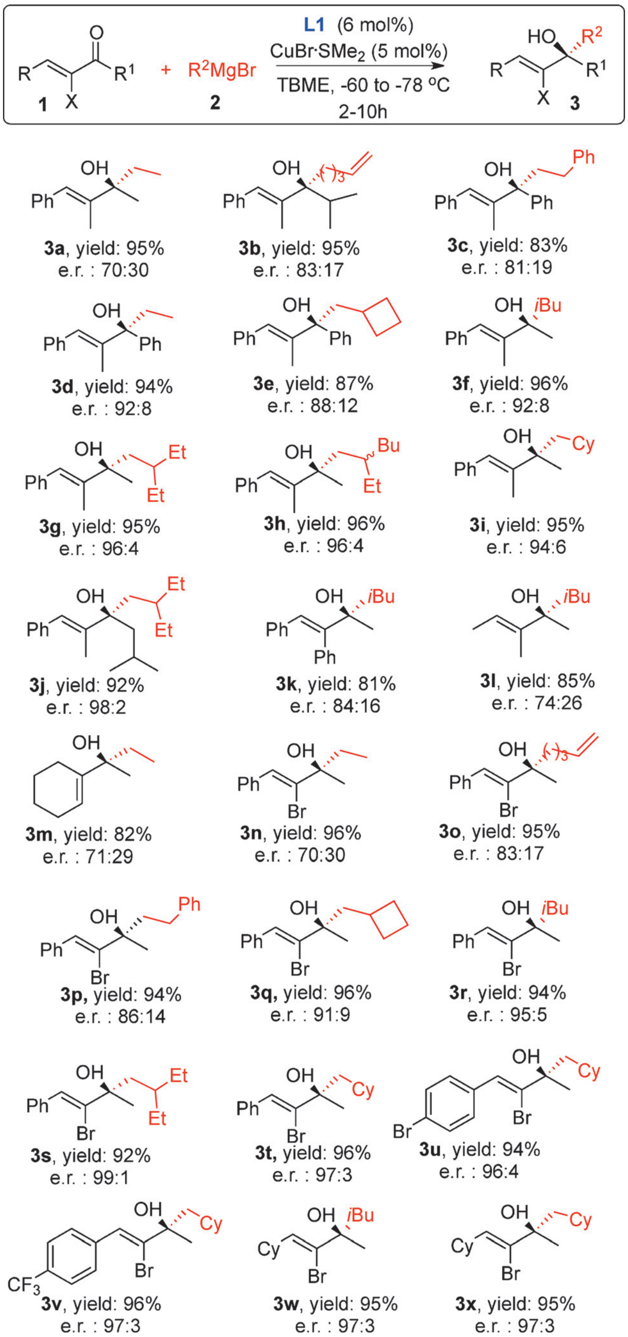
1,2- and 1,4-addition products were increased. Although the 1,4-addition product was still prevalent, the increased yield of the 1,2-addition product was a clear sign that Cu^{I} salt acts as a catalyst for its formation. Therefore it was envisioned that the balance might be steered towards 1,2-addition by combining Cu^{I} salt with the appropriate chiral ligand, which should also provide enantiodiscrimination.[16a]

An extensive chiral ligand screening[16a,b,18] led to the identification of ferrocenyl diphosphine **L1** (rev-JosiPhos) as a suitable ligand for asymmetric induction in the 1,2-addition product derived from *i*BuMgBr and ketone **1a**. Despite the still low regioselectivity, the allylic alcohol **3** was obtained with an e.r. of 64:36 (Scheme 3, entry 1). Other chiral ligands tested provided racemic 1,2-addition products. Subsequently, the less reactive α-Me-substituted α,β-unsaturated ketone **1b** was studied in the Cu^{I}-catalyzed addition of Grignard reagents. In the absence of both chiral ligand and Cu^{I} salt, mainly starting ketone **1b** remained. However, when Cu^{I} salt (but no chiral ligand) was added, both 1,2- and 1,4- addition products were formed in comparable amounts with approximately half of the starting ketone **1b** remaining. In the presence of both Cu^{I} salt and chiral ligand **L1**, a dramatic improvement in the reactivity of the ketone, 1,2-regioselectivity, and enantioselectivity was observed, providing the corresponding 1,2-addition product **3** with 97% yield and an e.r. of 91:9 (Scheme 3, entry 2). When a linear Grignard reagent (EtMgBr) was used in the addition to ketone **1b**, the product was obtained with excellent yield but lower e.r. (Scheme 3, entry 3).

Overall, these findings indicate that in the addition of Grignard reagents to α-H-substituted α,β-unsaturated ketones, regioselectivity is an important issue and that not only the α-substituent determines the regioselectivity but also the steric hindrance of the Grignard reagent is important for the stereoselectivity. These important initial results marked the beginning of the development of Cu^{I}-catalyzed synthesis of tertiary alcohols via addition of Grignard reagents to variety of ketones.

α-Substituted α,β-Unsaturated Ketones

Assessing the scope and generality of Cu^{I}-catalyzed 1,2-addition approach is crucial for its viability as a synthetic tool. The results for a large variety of Grignard reagents and substrates are summarized in Scheme 4.[16a,b] Various Grignard reagents provided excellent yields for 1,2-addition products, with branched Grignard reagents providing higher values in terms of stereoinduction (e.g., compare product **3a** with **3f** and **3g**). Products **3h** and **3i** were also obtained with excellent regioselectivity and isolated in yields of over 92% with high enantiomeric ratios. A diverse range of substituted α,β-unsaturated ketones was investigated. The presence of a phenyl group at the α′-position of the substrates led to good enantioselectivities, also with linear Grignard reagents such as $Ph(CH_2)_2MgBr$ and EtMgBr (Scheme 4; **3c,d**). A phenyl substituent at the α-position led to a slight decrease in selectivity (Scheme 4; **3k**). Use of the less reactive MeMgBr resulted in complete recovery of starting material, whereas addition of PhMgBr led to a racemic 1,2-addition product. Using a sterically hindered ketone, the



Scheme 4. Scope of products obtained from catalytic asymmetric addition of Grignard reagents to α-substituted α,β-unsaturated ketones.[16a,b] Yields refer to isolated products.

highly branched tertiary alcohol **3j** was obtained with an excellent yield and an enantiomeric ratio of 98:2. Ketones with lower inherent reactivities required higher reaction temperatures (−60 °C) to afford the 1,2-addition products **3l** and **3m** in high yields, albeit with lower enantioselectivities.

The scope was further expanded to α-brominated α,β-unsaturated ketones (Scheme 4, products **3n–x**).[16b] 1,2-Addition of branched Grignard reagents gave the corresponding tertiary alcohols **3r–t** in excellent yields and high enantioselectivities. An excellent enantiomeric ratio (99:1) and high yield of

3 s were also obtained when the reaction was scaled up to 3 mmol (Scheme 4). Addition of a Grignard reagent bearing a cyclobutyl moiety furnished the product **3 q** with an e.r. of 91:9 and a yield of 96 %. Butenyl- and phenylethylmagnesium bromide were also employed successfully, giving **3 o** and **3 p**, respectively. Notably, the reaction times with both α-Me- and α-Br- substituted ketones did not exceed 10 h overall and yields of isolated product exceeded 90 %.

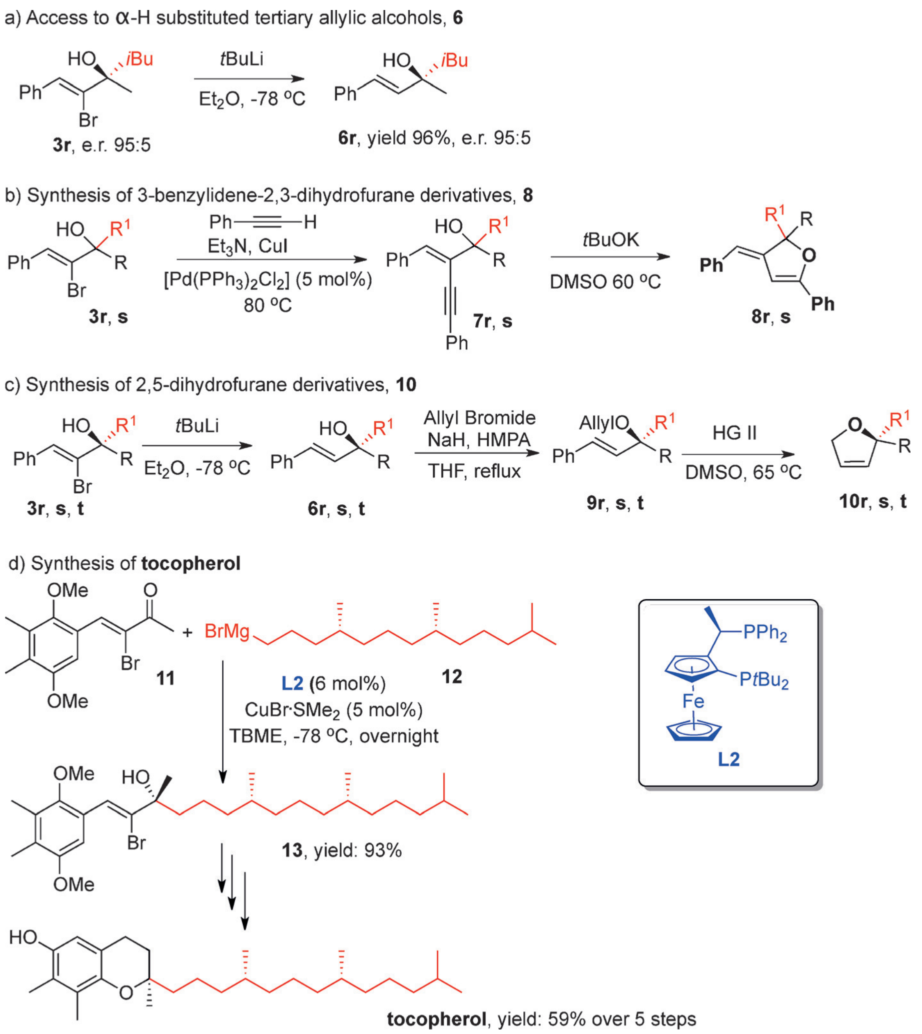
To access to α-H-substituted allylic alcohols, the α-Br-substituted alcohols **3 n–x**, obtained from α-Br substituted α,β-unsaturated ketones, can be easily debrominated (Scheme 5 a). For the 1,2-addition product **3 r**, this procedure afforded the corresponding product **6 r** with 92 % yield and retention of e.r. (98:2).[16b,19]

The current methodology was applied to the synthesis of chiral 2,3- and 2,5-dihydrofurans with tertiary oxygen-containing stereocenters (Scheme 5 b and c).[19] With the products **3** prepared by catalytic asymmetric 1,2-addition of Grignard reagents, Sonogashira reactions were carried out. The resultant corresponding enynes **7 r** and **s** were subsequently treated with *t*BuOK in DMSO to afford the desired 2,3-dihydrofurans **8 r** and **s** in good yields (Scheme 5 b).

Synthesis of the 2,5-dihydrofurans **10** also started from the chiral tertiary alcohols **3** (Scheme 5 c). Debromination with *t*BuLi yielded the desired alcohol products **6 r–t** in good isolated yields. Alkylation of the hydroxy group with allyl bromide furnished products **9 r–t** in high to excellent yields. Compounds **9 r–t** were subsequently transformed into the corresponding 2,5-dihydrofurans **10 r–t** by treatment with 5 mol % Hoveyda–Grubbs second-generation catalyst.

Finally, the copper-catalyzed 1,2-addition methodology was applied to the synthesis of *(R,R,R)*-γ-tocopherol (Scheme 5 d).[20] It was synthesized in 36 % yield over 12 steps (longest linear sequence). To construct the chiral center in the chroman ring, the 1,2-addition of a phytol-derived Grignard reagent to an α-Br-substituted ketone **11** prepared from 2,3-dimethylquinone was carried out. The best chiral ligand for this transformation turned out to be **L2**, providing the corresponding tertiary alcohol **13** with e.r. of 82:18 (Scheme 5 d).

With this scope and applications in hand, it was clear that the 1,2-addition methodology is indeed a viable, even promising, avenue for catalytic, asymmetric addition to α,β-unsaturated ketones. However, this was to be only a first stepping stone on the way to opening up new synthetic routes for several other important classes of substrates.

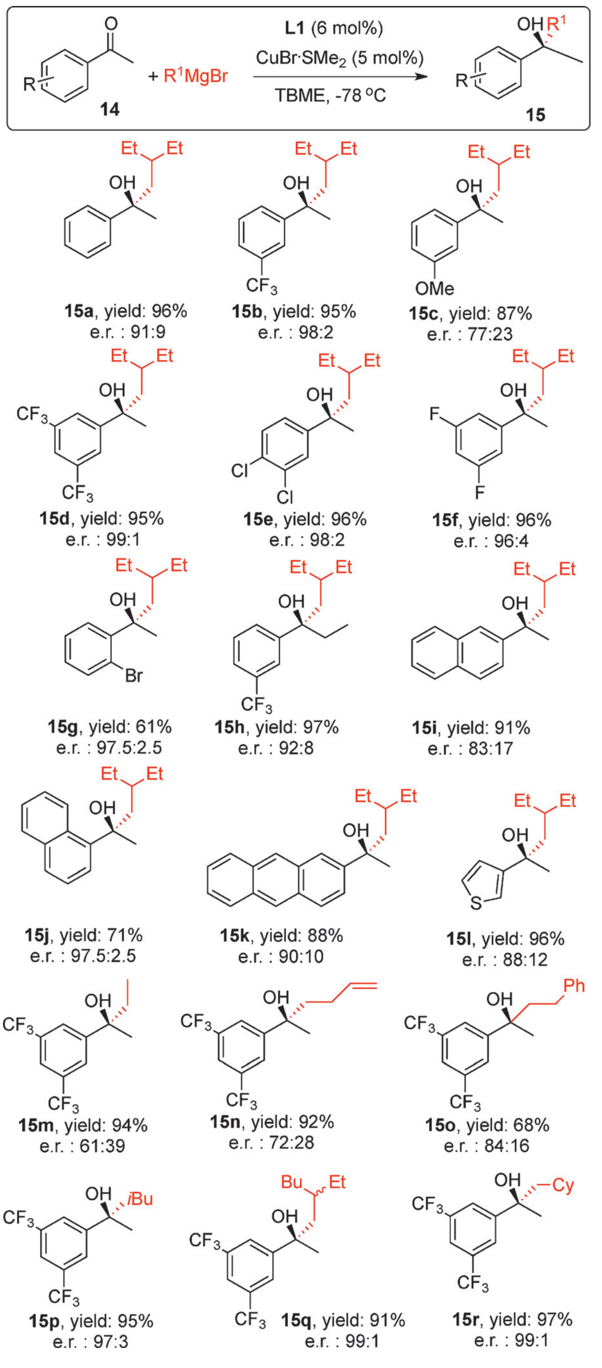


Scheme 5. Application of the methodology.[16b,19,20]

Aryl Alkyl Ketones

Tertiary alcohols derived from aryl alkyl ketones are common subunits in biologically active molecules and are important building blocks for natural product synthesis.[2] Therefore, methods for their catalytic enantioselective preparation are required. When compared to α,β-unsaturated ketones, aryl alkyl ketones are less reactive and enantiodiscrimination is also more difficult. Until recently, the catalytic methodologies for the alkylation of aryl alkyl ketones were based on the use of organozinc, organotitanium, and organoaluminum reagents,[3–8] whereas readily available Grignard reagents have only been used with superstoichiometric amounts of either additives[10] or a chiral ligand.[21] Even racemic synthesis of alcohols by direct addition of Grignard reagents to ketones was not known until Ishihara's recent development of zinc(II)-catalyzed non-asymmetric addition of Grignard reagents to ketones.[22]

Applying the Cu^{I}-based catalytic 1,2-addition system to the addition of Grignard reagents to aryl alkyl ketones proved a real game changer.[16c] Initial experiments were carried out using acetophenone as the substrate and 2-ethylbutylmagnesium bromide as the Grignard reagents in the presence of chiral Cu-catalyst in tert-butyl methyl ether (TBME; Scheme 6). Once again, chiral ligand screening singled out **L1** as the most optimal ligand in terms of both obtained yield (96%) and e.r. (91:9). Excellent results were obtained for a broad spectrum of substituted acetophenones (Scheme 6). For example product **15b** was obtained from 3-trifluoromethyl acetophenone with an excellent e.r. (98:2). Also remarkable are the excellent enantioselectivities obtained for products **15d–f**, derived from 3,5-di(trifluoromethyl) acetophenone, 3,4-dichloro acetophenone and 3,5-difluoro acetophenone, respectively. An *ortho*-Br substituent was also tolerated, although the yield of the product **15g** was decreased. It was also possible to synthesize product **15h**, derived from trifluoromethyl propiophenone, with an excellent yield and an e.r. of 92:8. A diminished enantioselectivity was obtained for product **15i**, but 1-acetonaphthone gave the product **15j** with an excellent e.r. (97.5:2.5). Heteroaromatic substituents were also tolerated, as in the case of product **15l** derived from 2-thiophenyl methyl ketone. Additions to alkyl alkyl ketones (not depicted) and the addition of phenylmagnesium bromide both afforded racemic products.



Scheme 6. Scope of products obtained from catalytic asymmetric addition of Grignard reagents to aryl alkyl ketones.[16c] Yields refer to isolated products.

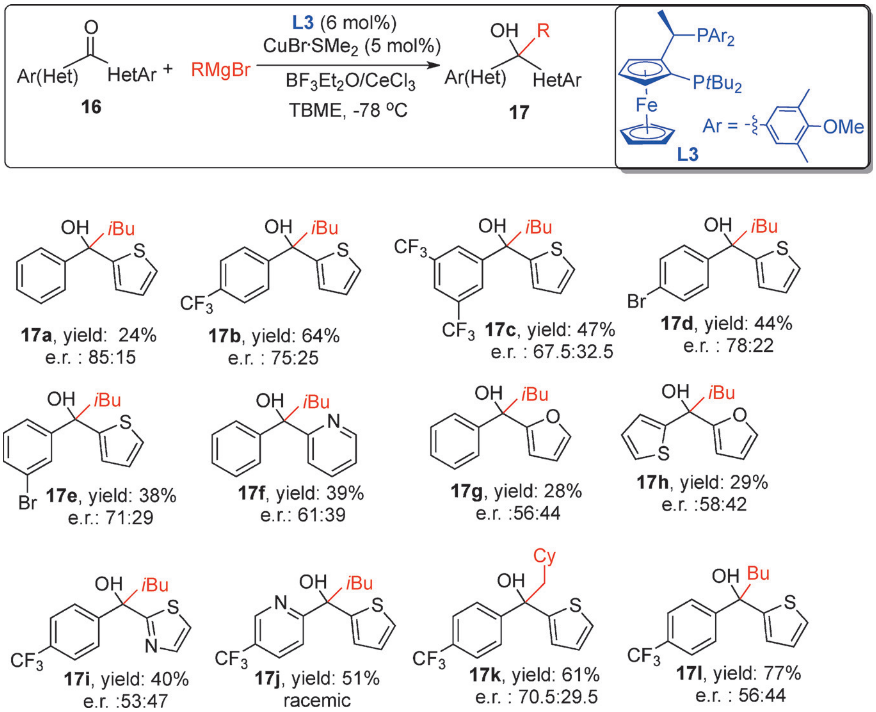
The scope of Grignard reagents studied in the addition to 3,5-di(trifluoromethyl) acetophenone showed a similar trend to that observed with α,β-unsaturated ketones: although all products **15m–r** were obtained with excellent yields, bulkier Grignard reagents, such as those with branched alkyl groups, were required to obtain high enantioselectivities. Methylmagnesium bromide was inactive and ethylmagnesium bromide gave product **15m** with only 61:39 e.r. With butenylmagnesium bromide, the e.r. of product **15n** rose to 72:28 and with phenylethylmagnesium bromide product **15o** was obtained with an e.r. of 84:16. The enantiomeric ratio of the product **15p** obtained from addition of isobutylmagnesium bromide reached 97:3. The additions of 2-ethylbutylmagnesium, 2-ethylhexylmagnesium, and cyclohexylmethylmagnesium bromides afforded the corresponding alcohols **15d**, **q**, and **r**, each in 99:1 e.r. (Scheme 6).

This study demonstrates that, despite the high reactivity and basicity of Grignard reagents, excellent yields and high enantioselectivities can be achieved in the addition reactions to enolizable ketones. The Cu^{I}-based catalytic system is capable of outcompeting several undesired reaction pathways and provides excellent stereocontrol and yields for a fairly broad scope of ketones and branched Grignard reagents.

Aryl Heteroaryl Ketones

Given the extensive presence of aryl heteroarylmethanol derivatives with a quaternary center in drugs,[23] the next target substrate class for application of the present catalytic methodology was aryl heteroaryl ketones. Conventional methodologies for the synthesis of tertiary diaryl alcohols rely on catalytic asymmetric arylations of corresponding aryl alkyl ketones with diphenylzinc,[4,6] phenylboronic acids,[24] or triphenylborane,[25] as well as titanates prepared in situ from triaryl aluminum[8a] or Grignard reagents.[10] Direct catalytic asymmetric alkylation of diarylketones has never been reported, because the reactivity of diaryl ketones is significantly diminished in comparison to alkyl aryl ketones. Furthermore, due to the negligible steric and electronic differences between the two-aryl substituents on the carbonyl moiety, enantiodiscrimination is also more difficult.

The initial investigations of the application of the 1,2-addition methodology focused on the addition of *i*BuMgBr to 2-benzoylthiophene **16**.[26] In the presence of catalytic amounts of $CuBr\cdot SMe_2$/**L1** in TBME, this led to the formation of racemic secondary alcohols (Scheme 7) by non-catalytic Meerwein–Ponndorf–Verley-type (MPV) reduction, a typical side reaction occurring with unreactive ketones. The combination of Lewis acids $BF_3\cdot OEt_2/CeCl_3$ (1:1)[27] was required to improve the reactivity and to outcompete the non-catalytic MPV reduction. Despite the use of Lewis acids, a low conversion and a mixture of the addition product **17a** and the reduction side product were obtained. Using chiral ligand **L3** led to complete selectivity toward the addition product with increased e.r. Unfortunately, the optimization of reaction conditions showed that increasing the conversion led to the formation of the elimination side product by dehydration, which takes place both during the reaction and the purification.



Scheme 7. Scope of products obtained from catalytic asymmetric addition of Grignard reagents to aryl heteroaryl ketones.[26] Yields refer to isolated products.

For the substrate scope, ketones with electron-donating and -withdrawing groups were tested. With activated ketones, the amount of addition products **17b** and **17c** increased, albeit at the cost of enantioselectivity. The presence of electron-donating groups, such as *m*-methyl or *o*-methoxy, in the phenyl ring led to almost no conversion. Ketones with bromine in the *para* or *meta* position gave rise to the corresponding tertiary aryl heteroaryl methanols **17d** and **17e** in moderate yields and enantioselectivities (Scheme 7). Other heteroaromatic rings were also tested and 2-benzopyridine was found to be a suitable substrate for this reaction. A tertiary alcohol **17f**, derived from ketone with a pyridine ring, was isolated in 39% yield, but with low enantiomeric ratio. The addition to the furan bearing ketone was highly problematic, although the results slightly improved when $BF_3\cdot OEt_2$ was not added, allowing us to obtain the very unstable product **17g**. Similar results were obtained for product **17h** derived from a diheteroaryl ketone bearing both thiophene and furan rings. A thiazole moiety was incorporated in product **17i**, but low e.r. was obtained. Within this framework, β-branched Grignard reagents are required for asymmetric addition (products **17a–k**). When a linear Grignard reagent was tested, product **17l** was obtained with very low enantiomeric ratio, albeit with a good yield.

The low to moderate enantiomeric ratios obtained for addition to aryl heteroaryl ketones can be explained by increasingly smaller steric and electronic differences between the two substituents at the carbonyl group going from aldehydes to aryl alkyl ketones and, finally, to diaryl or dialkyl ketones. In this context, the e.r of 85:15 obtained for product **17a** is quite remarkable. Although relatively low yields and enantiomeric ratios were obtained, this is the first example of the catalytic asymmetric addition of organometallic reagents to stereochemically very challenging diaryl ketones.

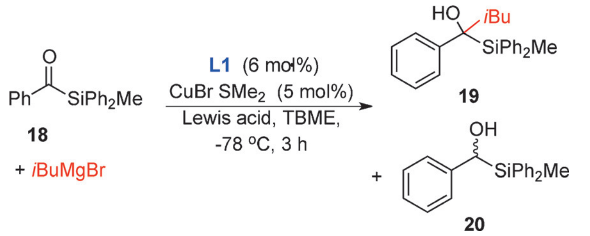
Acylsilanes

The development of synthetic methods for the enantioselective preparation of chiral silylated alcohols, which have important applications as silicon isosteres in medicinal chemistry due to their low toxicity and favorable metabolic profiles, is another important frontier in synthetic and medicinal chemistry.[28] Several strategies exist to access enantiopure secondary α-silylated alcohols,[28] but a general catalytic methodology to synthesize α-silylated alcohols with tetrasubstituted chiral carbon was unknown until recently.[29] The most direct approach to their synthesis is asymmetric addition of organometallic reagents to acylsilanes. The main obstacle for this transformation, as in the case of addition to diaryl ketones, is the non-catalytic MPV-type reduction, which provides racemic *secondary* α-silylated alcohols.

The Cu^{I}-catalyzed Grignard addition methodology was applied to acylsilanes to tackle the synthesis of chiral tertiary α-silylated alcohols.[27] The success of this approach is not obvious, considering the challenges that need to be overcome, including the bulkiness of the silyl group, the reduction of the carbonyl moiety, and a Brook rearrangement of the final alkoxide product.

The Cu^{I}/**L1** catalytic system proved once again to be the most efficient for this transformation. The desired α-silylated alcohol **19** was obtained with an e.r. of 95:5 (Table 1, entry 1).

Table 1. Optimization data for alkylation of acylsilanes with Grignard reagents.[27]

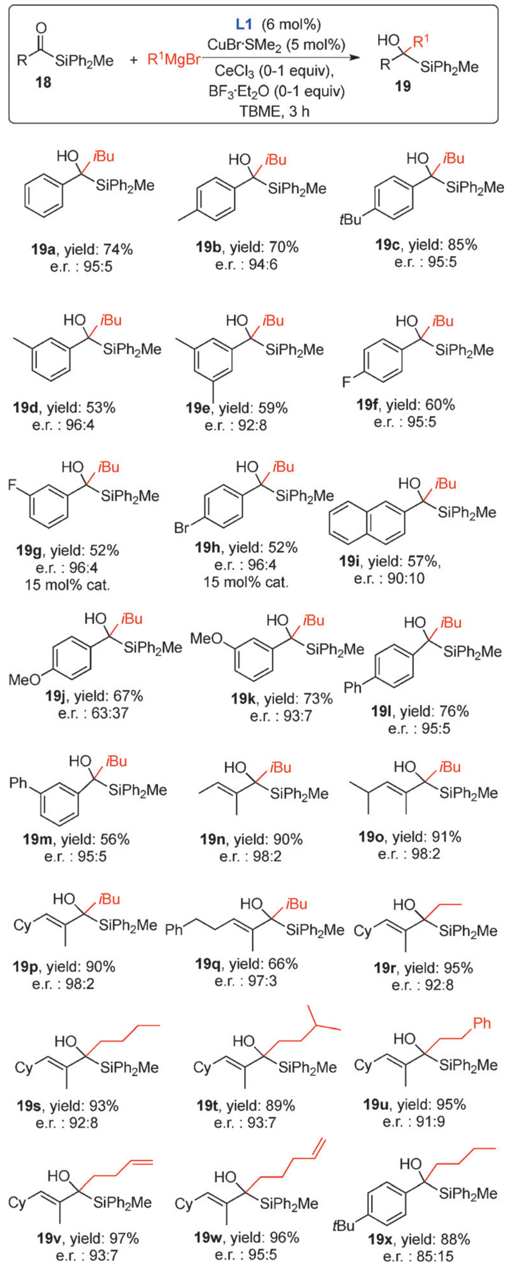


Entry	Lewis acid (equiv)	19/20	e.r. of 19 [%]
1	none	1:2	95:5
2	$CeCl_3$ (1.3)	1:1.5	96:4
3	TMSCl (2)	1:1.2	95.5:4.5
4	$BF_3 \cdot Et_2O$ (2)	3:1	93:7
5	$BF_3 \cdot Et_2O/CeCl_3$ (1:1)	5:1	95:5

As expected, the selectivity of the reaction was low (a 1:2 ratio of addition products to reduction products). Products arising from the Brook rearrangement were not observed. The reduction was addressed through variation of several parameters including the chiral ligands, solvents (Et_2O, THF), temperature (−100 °C, −30 °C) and substituents at the silyl atom ($SiPhMe_2$, $SiPh_3$ and $SiEt_3$ moieties) used. However, no combination was found that significantly improved the selectivity.

The reduction was presumed to be the result of activation of the carbonyl moiety of the acylsilane through coordination with the magnesium ion of the Grignard reagent, followed by β-hydride transfer. Hypothesizing that Lewis acid additives might prevent the coordination of the magnesium to the carbonyl moiety, thereby allowing the catalytic pathway to outcompete reduction, a screening of Lewis acids was performed. A 1:1 mixture of $BF_3 \cdot Et_2O$ and $CeCl_3$ (Table 1, entry 5) was identified as the key to achieving a near complete suppression of reduction.

With optimized reaction conditions in hand, the acylsilane scope was explored (Scheme 8). Tertiary α-silylated alcohols **19a**–**e**, derived from aromatic acylsilanes bearing alkyl substituents in the *para* or *meta* positions, were obtained with good yields and enantiomeric ratios of up to 96:4. Acylsilanes bearing halogens in the aromatic ring led to the corresponding tertiary alcohols **19f**–**h** with high levels of stereoselectivity, albeit with decreased selectivity of the alkylation. The selectivity of the reaction with MeO as substituent on the aromatic ring was highly dependent on its position (Scheme 8; **19j,k**): with *p*-OMe the e.r. was only 63:37, whereas with *m*-OMe an e.r. of 93:7 was obtained. Due to steric hindrance, neither alkylation nor reduction was observed with the *o*-OMe substituted substrate. Introducing phenyl substituents in the *para* or *meta*



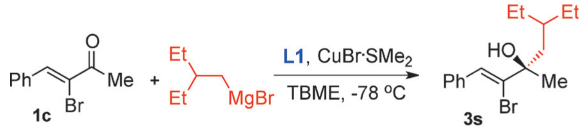
Scheme 8. Scope of products obtained from catalytic asymmetric alkylation of acylsilanes with Grignard reagents.[27] Yields refer to isolated products.

position of the aromatic ring furnished the corresponding tertiary α-silylated alcohols **19l** and **m**, both with e.r. values of 95:5. With aliphatic acylsilanes, only the reduction products were obtained.

The scope study also indicated that α,β-unsaturated acylsilanes are good substrates. The corresponding tertiary allylic α-silylated alcohols **19n**–**q** were obtained with excellent yields and enantiomeric ratios (Scheme 8). Since the MPV reduction was less pronounced for these substrates, the amount of the Lewis acid mixture was reduced to 0.25 equivalents.

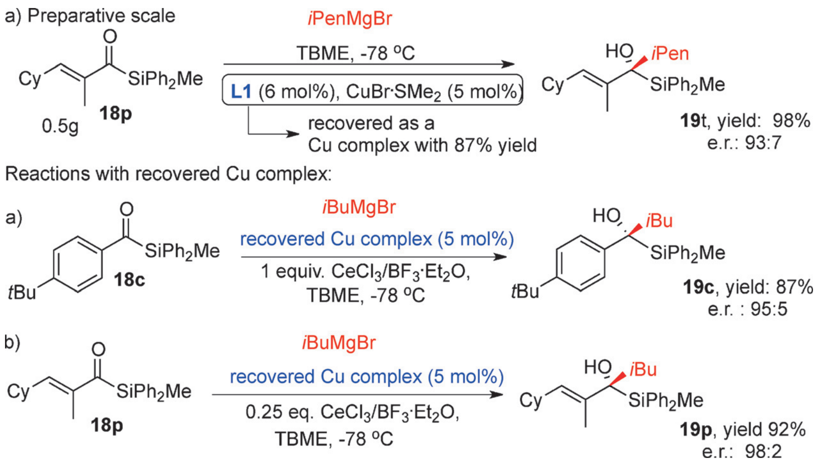
The scope of Grignard reagents was found to be very general, especially when compared to the classes of substrates discussed above (α,β-unsaturated ketones, aryl alkyl ketones, and diaryl ketones). The catalytic system can tolerate linear and β-branched, as well as functionalized Grignard reagents. In all cases, high yields and good to excellent stereoselectivities were obtained for the corresponding tertiary alcohols **19 r–x** (Scheme 8). A low conversion and a racemic product were obtained only with MeMgBr.

To demonstrate the synthetic potential of this reaction, it was carried out on a large scale (500 mg of **18 p**; Scheme 9 a); product **19 t** was obtained in an excellent 98 % yield with an e.r. of 93:7. Importantly, the catalyst could be also recovered by column chromatography as the Cu complex with 87 % yield and used repeatedly without any loss of catalytic activity (Scheme 9 b, c).



e.r. **L1**	e.r. **3s**
50 : 50	50 : 50
60 : 40	97 : 3
70 : 30	98 : 2
80 : 20	97 : 3
90 : 10	98 : 2
100 : 0	99 : 1

Scheme 10. Asymmetric amplification in 1,2-addition of Grignard reagent to α,β-unsaturated ketone **1 c** catalyzed by scalemic **L1**–Cu complexes.[31]



Scheme 9. Practical aspects. Yields refer to isolated products.

Asymmetric Amplification in Addition of Grignard Reagents to Ketones

A remarkable feature of the $CuBr{\cdot}SMe_2$/**L1** catalytic system is the presence of a dramatic asymmetric amplification,[30] as exemplified by the 1,2-addition of (2-ethylbutyl)magnesium bromide to ketone **1 c** (Scheme 10).[31] In practical terms, this means that even when the chiral ligand has an e.r. as low as 60:40, the highest level of asymmetric induction (e.r. = 97:3) is still reached.

This phenomenon was found to solely originate from a large difference in the solubility of the Cu complexes formed in the reaction mixture. The solubility of the heterochiral dimer was much lower than that of the homochiral dimer, thus leading to precipitation of the former and enhancing the reaction with the homochiral complex. Since this phenomenon is inherent to the Cu complex in TBME and independent of the reaction at hand, it can be applied to any type of reaction using this copper complex.

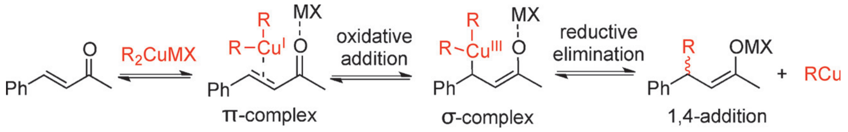
To assess whether Cu complexation is required for this phenomenon to occur, scalemic chiral ligand **L1** with different enantiomeric ratios was prepared and stirred for 24 h in TBME both at room temperature and at 0 °C. No precipitate was formed, clearly indicating that it is the metal complexation that changes the solid-solution phase behavior of this chiral diphosphine ligand in a given solvent. Follow-up research showed that this phenomenon is more general: complexation of other transition metals (Pd, Rh) with a number of chiral ferrocenyl diphosphine ligands, as well as with BINAP, led to extreme differences in solubility between the enantiopure and the racemic complexes.[32]

Mechanistic Considerations

In the early 20th century, seminal work by Ullman and Gilman[13b–d] laid the groundwork for modern organocopper chemistry and since then copper-mediated cross-couplings and organocuprate additions have become major tools in organic synthesis.[1–4] Nevertheless, organocopper chemistry has long remained one of the least mechanically understood synthetic methods.

Since the work by Kharash and Tawney in 1941,[13] it is known that if organometallics are used in combination with copper salts, or if instead the organocopper reagent is used in addition to α,β-unsaturated carbonyl compounds, 1,4-addition (conjugate addition) occurs predominantly. Cu^{I}/Cu^{III} redox chemistry has long been proposed to have a role in these reactions, but it took many years of experimental and computational chemistry before this hypothesis was supported by the groundbreaking experimental observation of the putative Cu^{III} intermediate in non-catalytic systems by using NMR spectroscopy.[32–34] The current mechanistic view for stoichiometric addition of organocuprates to α,β-unsaturated carbonyl compounds involves reversible formation of a Cu^{I}–olefin π-complex, followed by a formal oxidative addition to the β-carbon leading to a Cu^{III} intermediate and, finally, reductive elimination to form the enolate of the 1,4-addition product (Scheme 11).

Stoichiometric cuprate addition and copper-catalyzed asymmetric addition are generally assumed to share the same



Scheme 11. Mechanistic rationale proposed for stoichiometric 1,4-addition of cuprates.

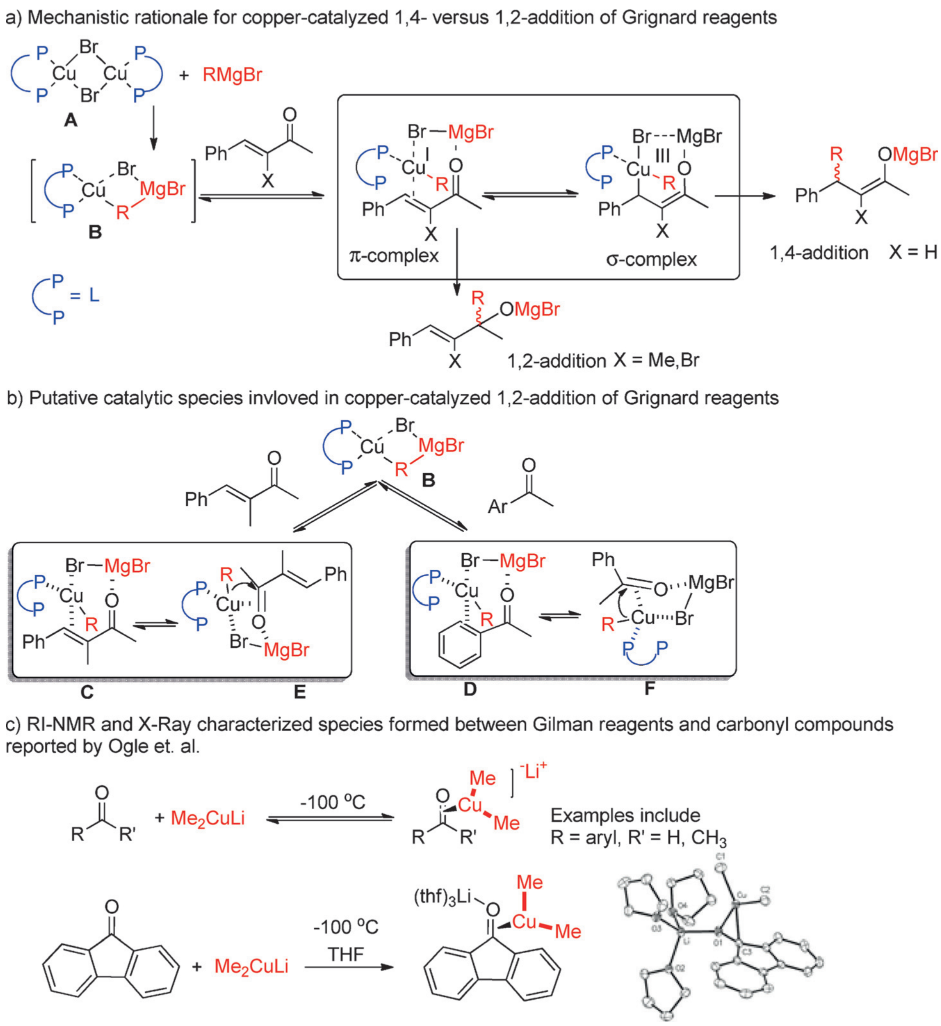
mechanistic principles, but no direct experimental evidence, including the observation of Cu^{I}/Cu^{III} redox chemistry for the catalytic reaction, has been reported to date.[12,14,32–34h]

In earlier studies, Harutyunyan and co-workers identified a mechanistic scheme for the copper catalyzed enantioselective 1,4-addition of Grignard reagents to α,β-unsaturated ketones and esters (Scheme 12a).[35] The proposed pathway of the reaction involves the formation of the catalytically active monomeric complex **B**, formed from dimeric precatalyst **A** by transmetalation with a Grignard reagent. Both complexes **A** and **B** were identified through the combination of spectroscopic studies. The central hypothesis is that the monomeric complex **B** functions in a similar manner to organocuprates, with the additional advantage that it bears a chiral diphosphine ligand. This accelerates the addition process and stabilizes the reaction intermediates (Cu^{I} π-complex and Cu^{III} σ-complex), thus overcoming background 1,4- and 1,2-addition reactions (Scheme 12).

From a mechanistic point of view, the preference for 1,2-addition brought about by similar 1,4-addition copper catalysts, as described in this concept review, is an enigma. The Cu^{I}-catalyzed 1,2-addition of Grignard reagents to α,β-unsaturated ketones, as well as alkylations of other ketones, described above cannot be explained by the existing mechanistic rationale involving Cu^{I}/Cu^{III} redox chemistry. A major requirement to persuade the copper catalyst to deliver the nucleophile at the carbonyl (1,2-addition) is the presence of an α-substituent in the α,β-unsaturated ketone and a *trans* relationship between the carbonyl group and the β-substituent around the double bond.

Based on experimental findings for Cu^{I}-catalyzed 1,4-additions,[32,33] one can expect that the initial stage of the catalytic cycle for 1,2-addition to α,β-unsaturated ketones also involves formation of monomeric complex **B** from dimeric precatalyst **A**, followed by reversible formation of the π-complex with the ketone (Scheme 12a). Typically, oxidative addition to form a Cu^{III} σ-complex must be the next step, but the equilibrium constant between the π- and σ-complexes will depend on the stability of the Cu^{III} intermediate. Most likely, the presence of an α-substituent in the α,β-unsaturated ketone destabilizes and prevents accumulation of the Cu^{III} species, which in turn prevents 1,4-addition and favors accumulation of the π-complex, followed directly by 1,2-addition (Scheme 12a).

Although this explanation for 1,2-selectivity might sound reasonable, it should be kept in mind that an identical catalytic system is required for 1,2-addition to both α,β-unsaturated and aryl alkyl ketones. Thus, it is more plausible that after π-



Scheme 12. Mechanistic considerations for asymmetric Cu^{I}-directed 1,4- versus 1,2-addition of Grignard reagents to carbonyl compounds.

complexation with the conjugated double bond (Scheme 12 b; α,β-unsaturated ketone, species **C**) or with the aromatic ring (Scheme 12 b; aryl alkyl ketones, species **D**) Cu migrates intramolecularly from the double bond to the carbonyl moiety to form a new π-complex (Scheme 12 b, species **E** and **F**). The coordination to the carbonyl double bond might also occur directly, without initial π-complexation with the conjugated double bond or aromatic ring. This coordination mode provides double activation of the substrate via a pseudo-chair transition state: Lewis acid activation of the carbonyl moiety by Mg^{2+}, and activation of the carbonyl double bond by copper (Scheme 12 b) in analogy with the coordination mode of organocopper species reported by Ogle and co-workers[34i,j] (Scheme 12 c). Finally, formation of the π-complex **E** or **F** is followed by an alkylation step to form the product alkoxide (Scheme 12 b).

Conclusion and Outlook

Enabling the utilization of the highly reactive Grignard reagents for the catalytic synthesis of chiral tertiary alcohols is an important advance in terms of cost, efficiency, and sustainability. In this Concept, we have described how the discovery of the application of Cu^I complexes for this purpose came about, and how tuning of the reactivity through appropriate chiral ferrocenyl-based diphosphine ligands has established a wide scope in both Grignard reagents and substrates. The resulting catalytic systems can for the first time compete effectively with the uncatalyzed reactions of organomagnesium reagents and the alternative reaction pathways. Crucially, this novel methodology allows the unprecedented direct use of Grignard reagents in catalytic asymmetric alkylations of carbonyl compounds. This provides several important benefits, including significantly reduced reaction times, a wide substrate scope comprising α-substituted α,β-unsaturated ketones, aromatic ketones, and acylsilanes and, finally, access to highly valuable chiral tertiary allylic and benzylic alcohols with high yields and enantiomeric ratios. More than this, however, the methodology breaks a 70-years-old paradigm in organic chemistry that reflects the widespread acceptance that asymmetric 1,4-addition of Grignard reagents to α,β-unsaturated ketones requires Cu^I-based catalysts to outcompete 1,2-addition. The results described herein clearly demonstrate that Cu^I-based complexes can be very efficient catalysts for both direct alkylation of ketones in general and 1,2-additions to α,β-unsaturated ketones using Grignard reagents.

Thus, a major step forward has been made in catalytic asymmetric synthesis of chiral tertiary alcohols through addition of organometallic reagents. However, further research is required to improve the control over the regioselectivity in additions of Grignard reagents to α-H-substituted α,β-unsaturated ketones and over stereoselectivities in the addition of methyl, phenyl, and linear Grignard reagents. Furthermore, additions to dialkyl and diaryl ketones are still a major challenge when Grignard reagents or other organometallics are considered. Improvements in chiral ligand design need to be made in the years ahead in order to meet these challenges. Alongside synthetic efforts exploring the scope of the reaction, this catalytic system would greatly benefit from detailed mechanistic work focused on elucidating each step in the catalytic cycle and the species involved. Finally, one can imagine that this Cu^I-based catalytic system has the potential to address major problems in synthetic organic chemistry, such as catalytic enantioselective additions of organometallics to less reactive ketimines to furnish the corresponding highly valuable chiral-carbon tertiary amines. Overall the methodologies described herein have the promise to open up new synthetic routes for organic chemists and signal the advent of an entirely new role for Cu^I-based catalysts in asymmetric synthesis.

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